PERFORMANCE EVALUATION REPORT

Accuracy, EER, F1-Score, and Confusion Matrix

1. Executive Summary

This performance report provides a rigorous quantitative evaluation of the deployed classification model. By assessing standard validation metrics alongside biometrics-centric benchmarks, this report establishes a holistic baseline of system reliability, discriminative power, and error distribution. The evaluated model demonstrates robust foundational stability, achieving high global alignment with ground-truth classifications while optimizing trade-offs between false acceptance and false rejection regimes.

2. Core Performance Metrics

The model's operational performance is summarized across four core pillars: Accuracy, Equal Error Rate (EER), Precision, and the F1-Score. Each metric exposes distinct behavioral characteristics of the system pipeline.

Metric	Value
Accuracy	**89.94%**
Equal Error Rate (EER)	**10.07%**
F1-Score	**90.15%**

3. Detailed Analytics & Interpretations

3.1 Accuracy

While Accuracy (89.94%) indicates a highly successful rate of correct classification, it can mask underlying vulnerabilities if the evaluation data exhibits heavy class imbalance. In this assessment, class distributions are relatively uniform, validating global accuracy as an authentic indicator of general system effectiveness.

3.2 Equal Error Rate (EER) Dynamics

The system's Equal Error Rate stands at 10.07%. The EER is critical for security, verification, and matching tasks. It denotes the precise decision boundary threshold where the False Acceptance Rate (FAR) and False Rejection Rate (FRR) converge. A lower EER highlights superior intra-class compactness and inter-class separation. At this operating point, the model minimizes risk evenly between unauthorized intrusions and false alarms.

3.3 F1-Score Assessment

The F1-Score serves as a stricter measurement than accuracy, as it forces an equilibrium between Precision (minimizing false positives) and Recall (minimizing false negatives). The marginal delta between the Macro and Weighted F1-scores indicates that system precision remains highly resilient and does not collapse on smaller or less frequent target sub-populations.

4. Confusion Matrix Analysis